

DATE	XD12R TRIM JIG
03.13	LOGIC DESIGN START
03.12	BLOCK CHANGE
03.21	LED
03.22	PCB START
03.25	회로 REVIEW
04.04	TEST 별 CASE 1/2 적용
04.07	TEST OPTION CONNECTOR 추가
05.29	FPGA DAC 삭제
06.06	XD24R R0, R1, IC603
06.09	XC24R FB1-3 TO MPU

XC24R FB1-3 TO MPU

- 05.16
1. 기존 XD12R Trim 지그 보드 + 새로운 XC24R IC 추가 (첨부 Data sheet참조)

2. MCU Nucleo에서 새로운 XC24R IC Trimming 기능이 있고 지그 보드에서 XC24R 양품/불량 선별

3. XC24R은 총 3개이며 첫번째칩은 Custom socket IC이고 첫번째 칩과 daisy 연결해 2,3 번째 XC24R IC연결됨 (daisy연결은 data sheet참조)

4. XD12R 첫번째 칩에 제어신호는 PIANO SW 4개짜리로 연결함 (아래 그림 SW7, SW8제어신호에 PIANO 스위치 각각 연결)
(FPGA , MCU, XC24R, reserved 4개)

5. XC24R SVO 신호가 9개입니다.
이중 한 채널을 선택하여 XD12R 첫번째 칩에 연결합니다 (채널 선택은 PIANO SW 여러 개로 삽입해 그중 1개만 ON선택)

6. XC24R 출력 D가 24개입니다.
이중 한 채널을 선택하여 XD12R 첫번째 칩에 연결합니다 (채널 선택은 PIANO SW 여러 개로 삽입해 그중 1개만 ON선택)

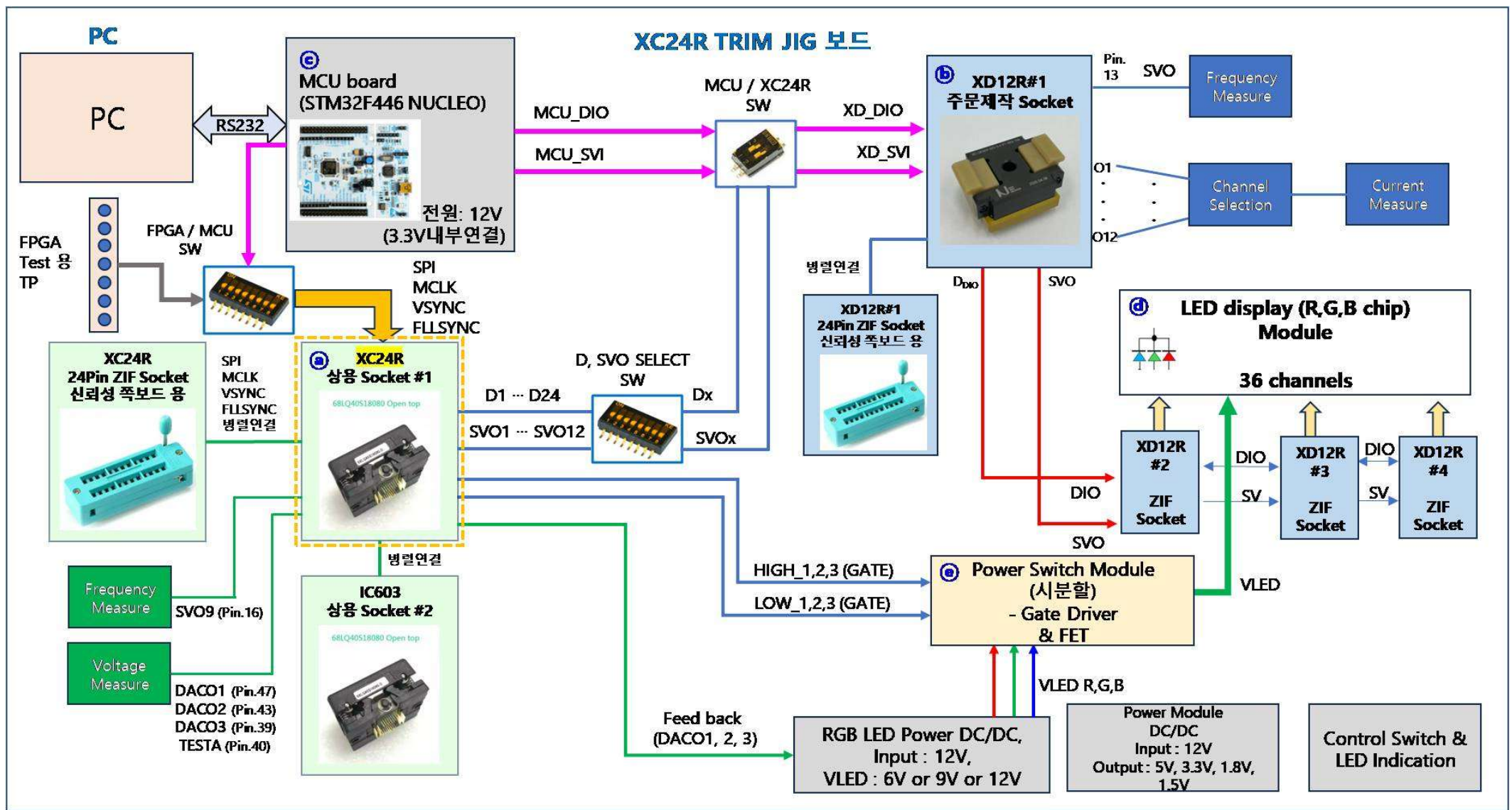
7. GATE DRIVER enable 신호가 FPGA에서 high, low 3개씩 각각 VLED power 제어신호로 받습니다.
지금 회로는 FPGA에서 단독으로 Gate IC에 연결되는데
PIANO 스위치로 FPGA 와 XC24R IC을 선택해서 GATE Driver chip으로 입력되게 합니다

8. VLED DAC FB 신호가 FPGA에서 받아서 제어합니다 (R,G,B)
지금 회로는 DAC IC 출력을 단독으로 받아서 FB 제어하는데
PIANO 스위치로 FPGA - DAC 과 XC24R IC을 선택해서 DC-DC Buck IC FB로 입력되게 합니다

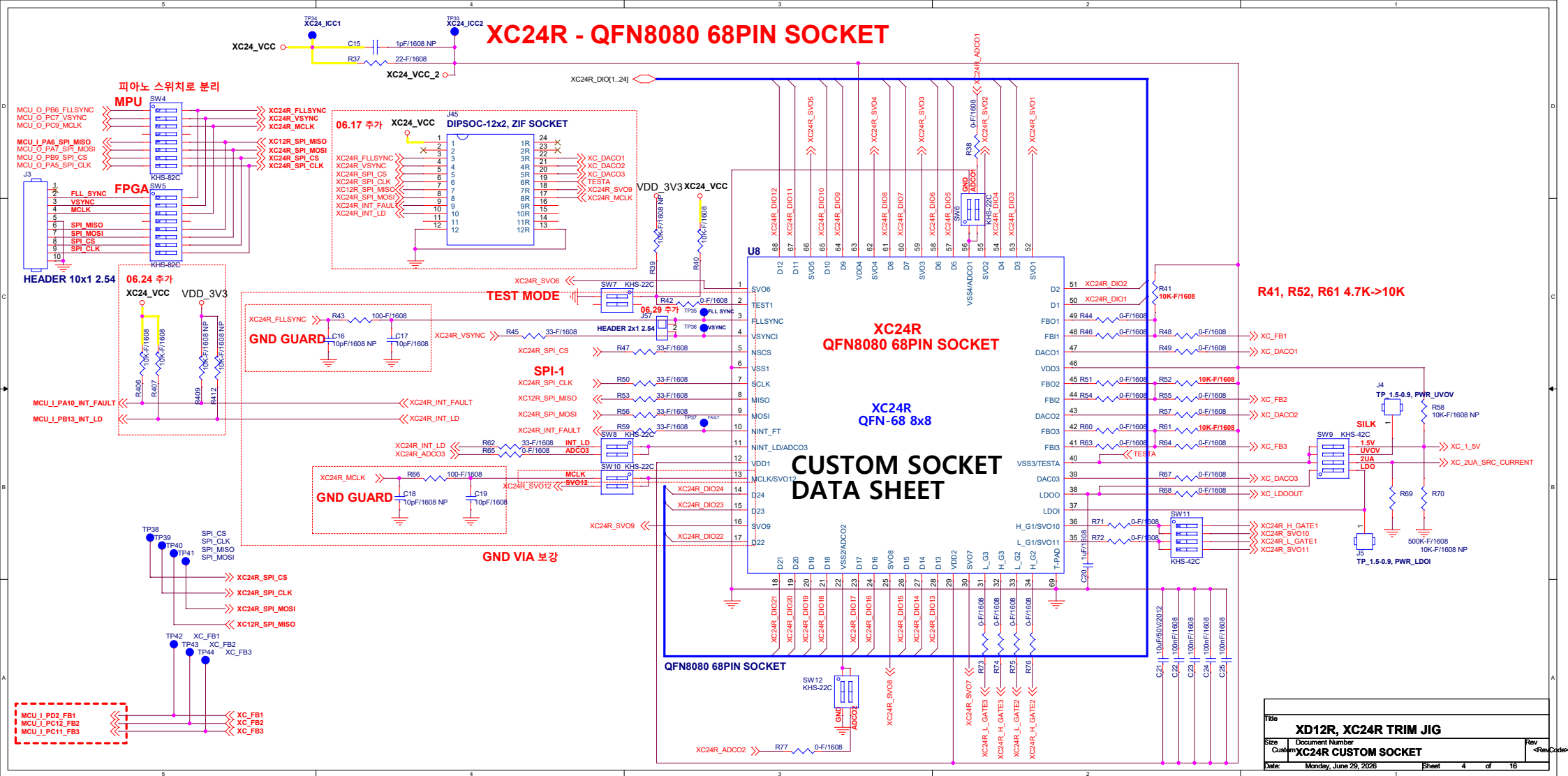
9. MCU에서 XC24R IC Trimming 제어하기 위해 필요한 GPIO 신호는
Data sheet Pin assign 확인해서 연결해주세요
Port 부족 시 확장 IC추가하고요

10. FPGA보드는 최종 XD12R 지그보드와 동일하게 합니다

Title		
XD12R, XC24R TRIM JIG		
Size	Document Number	Rev
Custom	REVISION HISTORY	1R0
Date:	Saturday, June 27, 2026	Sheet 1 of 16

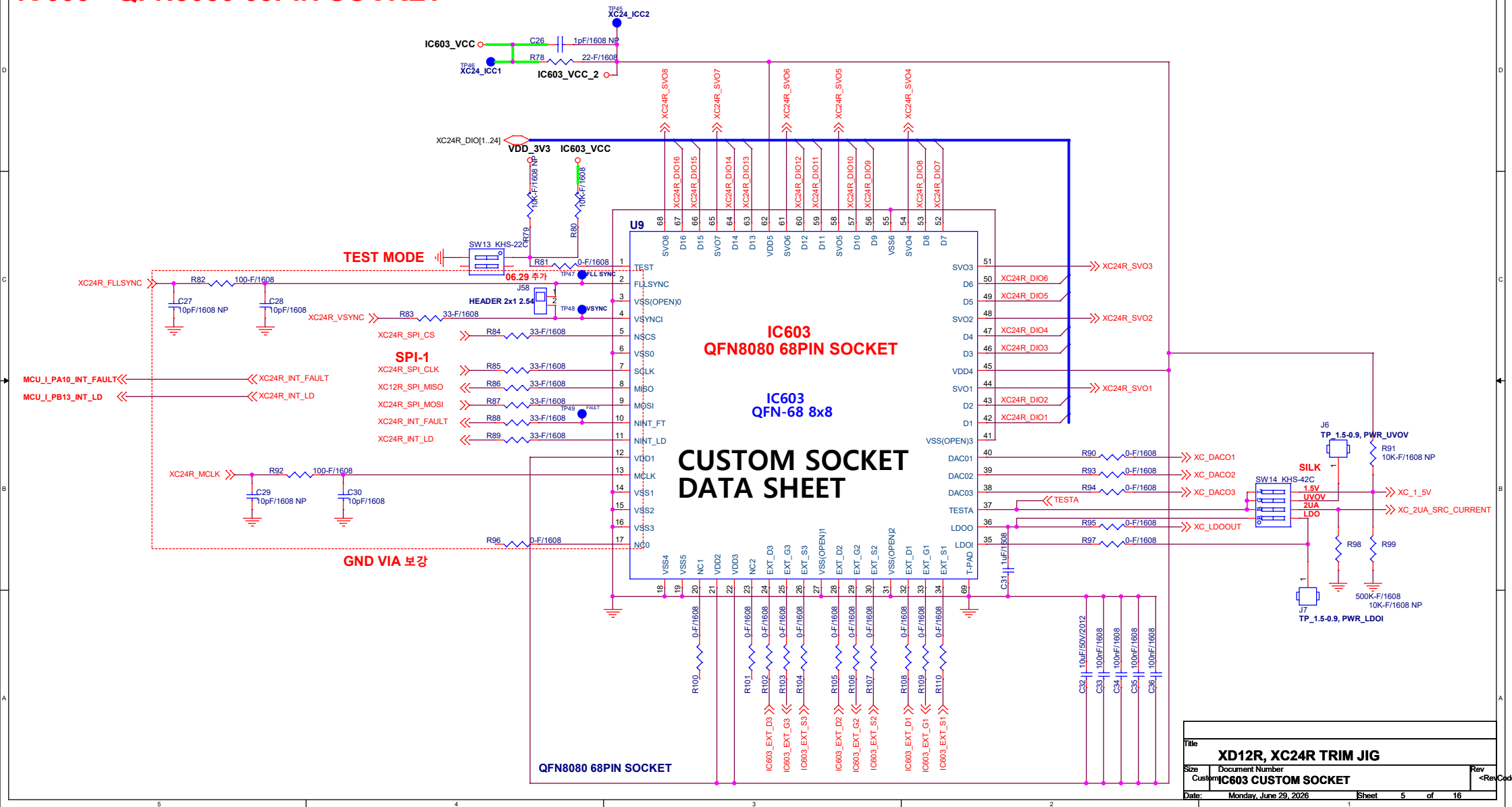


XC24R - QFN8080 68PIN SOCKET

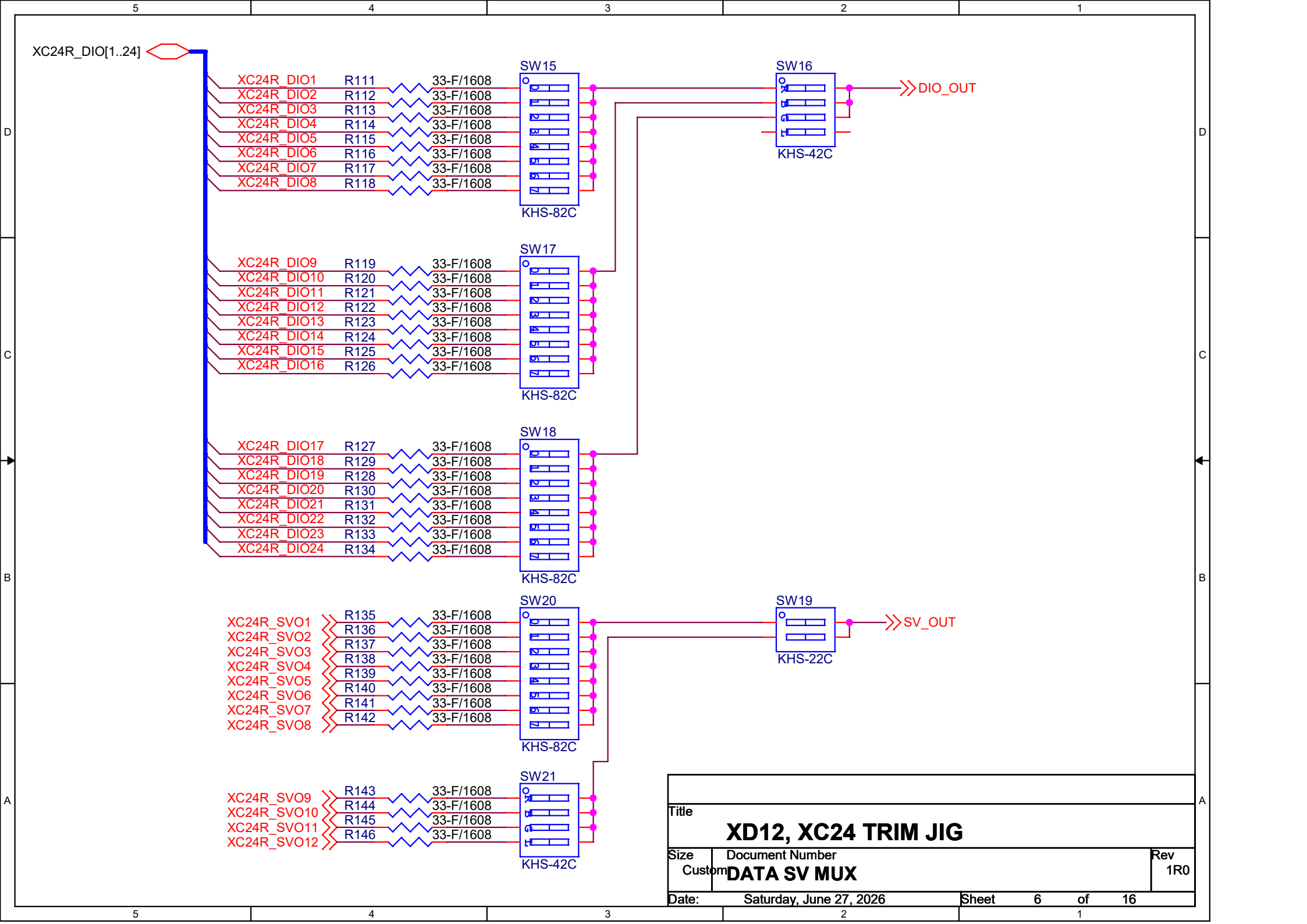


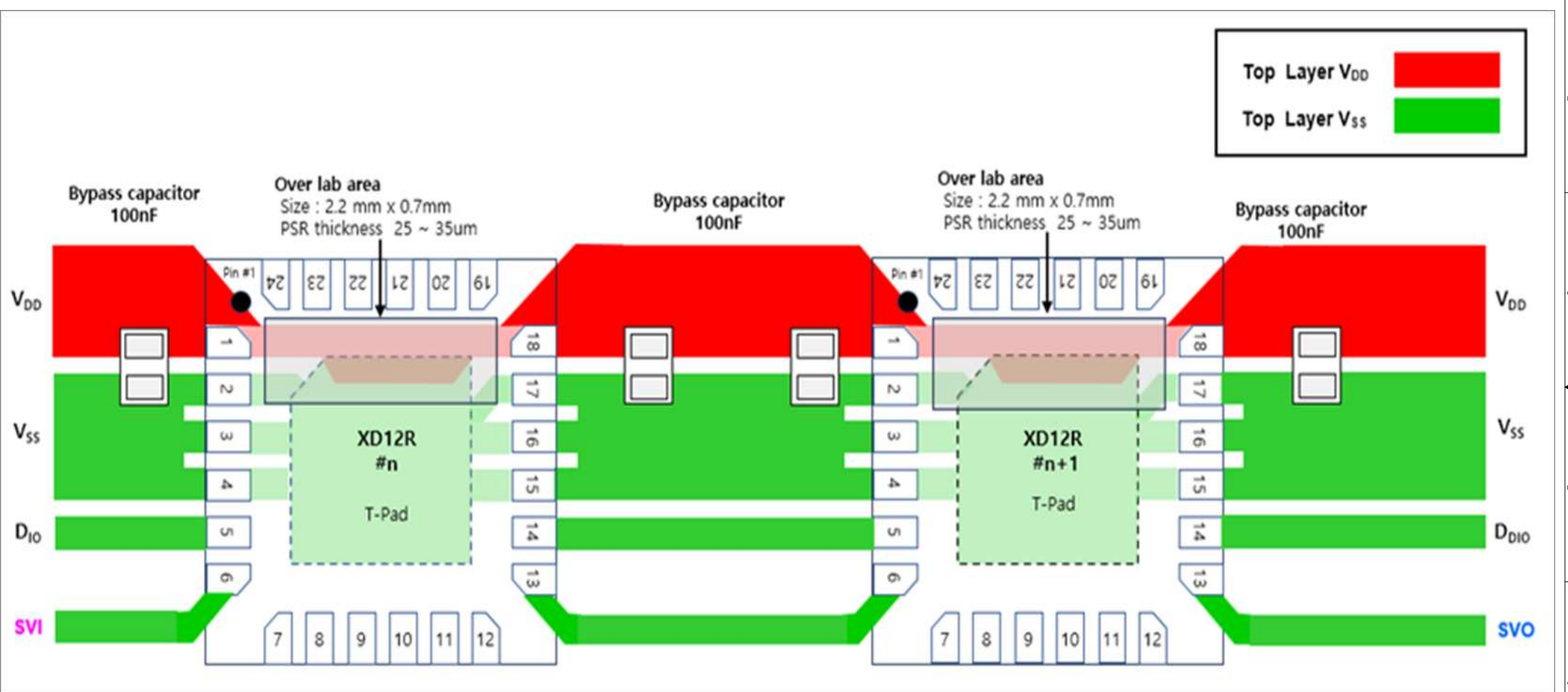
Title			XD12R, XC24R TRIM JIG		
Size			Document Number		
Customer			XC24R CUSTOM SOCKET		
Date			Monday, June 29, 2028		
			Sheet 4 of 16		

IC603 - QFN8080 68PIN SOCKET



Title			
XD12R, XC24R TRIM JIG			
Size	Document Number	Rev	<Rev Code>
Custom	C603 CUSTOM SOCKET		
Date:	Monday, June 29, 2026	Sheet	5 of 16



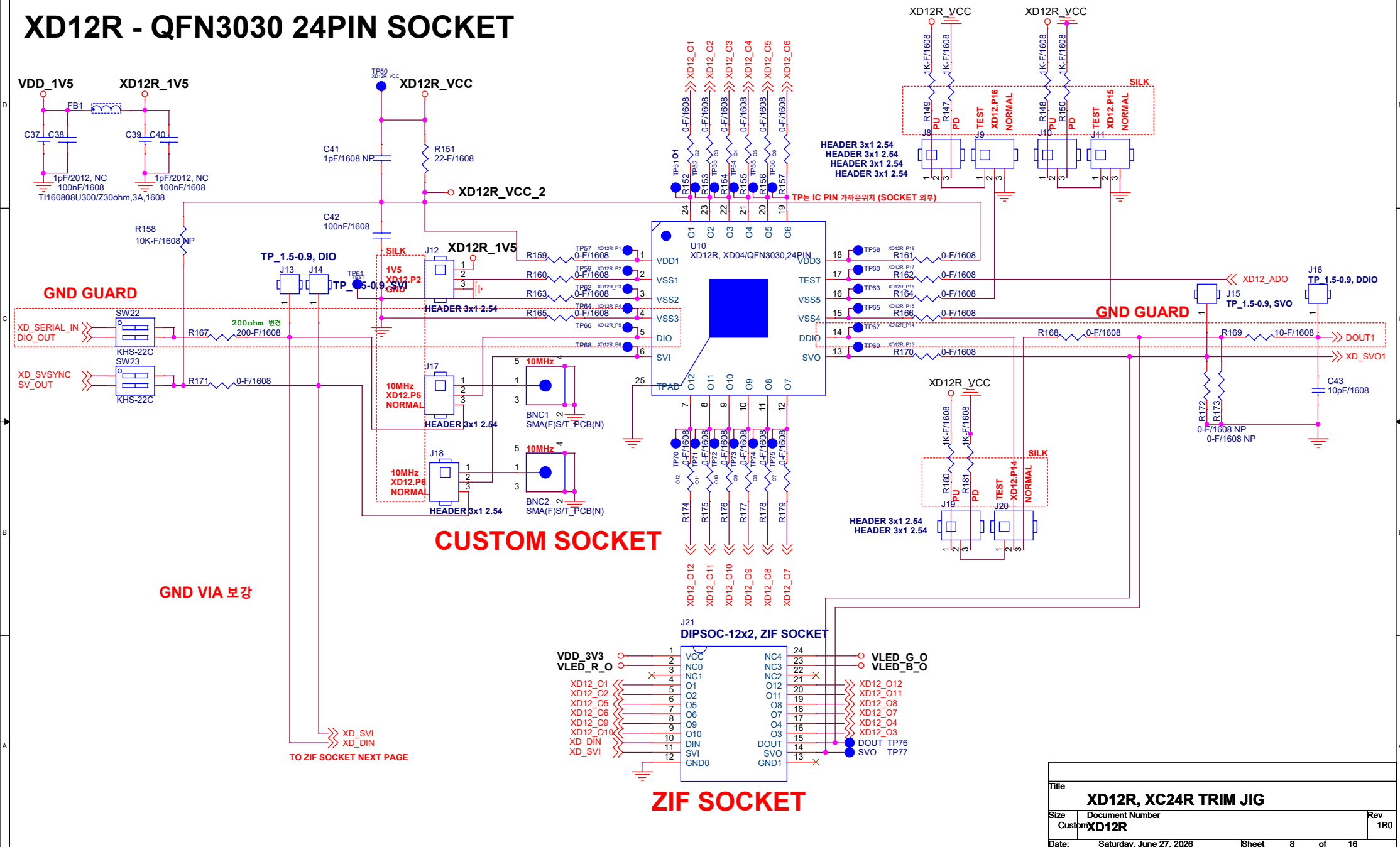


Top Layer V_{DD}

Top Layer V_{SS}

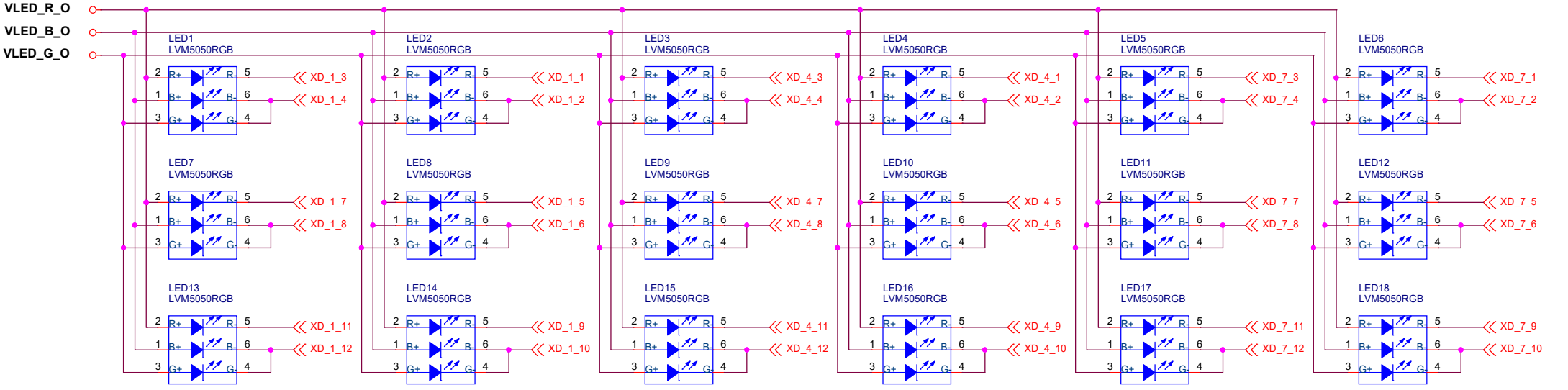
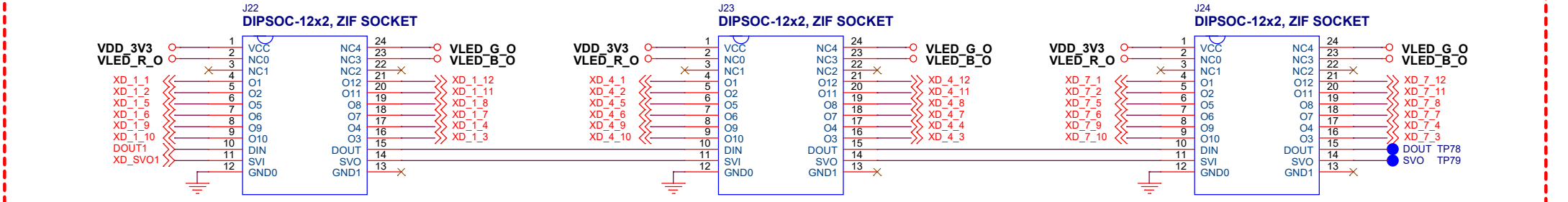
Title			XD12R, XC24R TRIM JIG	
Size			Document Number	
Cust			XD12R LAYOUT GUIDE	
Date			Saturday, June 27, 2026	Rev 1R0
			Sheet 7 of 16	

XD12R - QFN3030 24PIN SOCKET



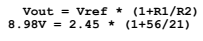
Title				
XD12R, XC24R TRIM JIG				
Size	Document Number			Rev
Custom	XD12R			1R0
Date:	Saturday, June 27, 2026	Sheet	8 of 16	

XD12R - ZIF SOCKET (ZIF SOCKET 간섭 고려)



Title		
XD12R, XC24R TRIM JIG		
Size	Document Number	Rev
Custom	XD12R RGB LED	1R0
Date:	Saturday, June 27, 2026	Sheet 9 of 16

```
"L" = VLED_9V -> 9V
```

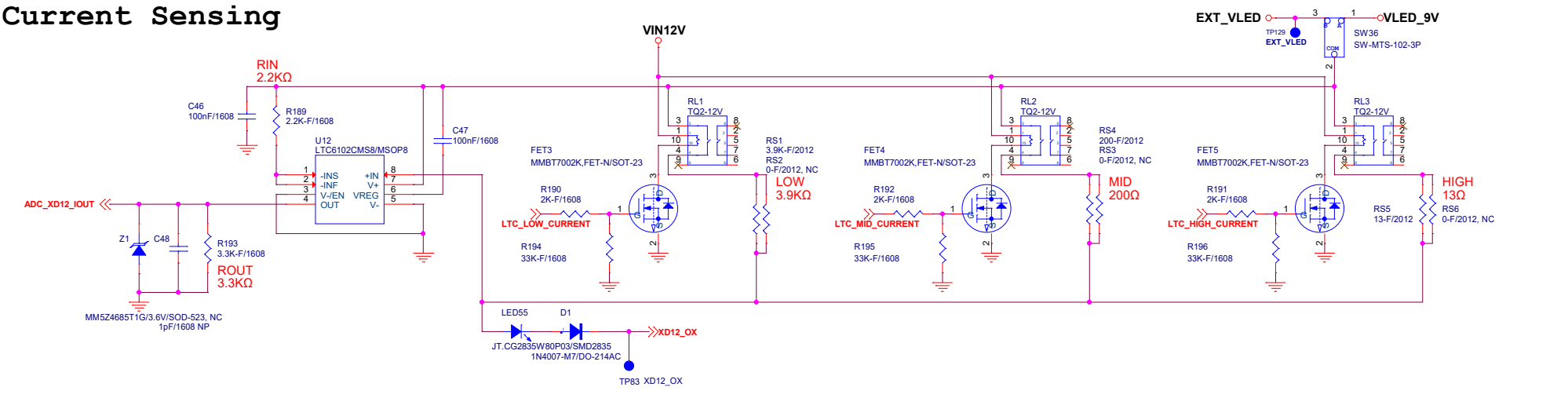


RED Vf=2.0 If=20mA
GREEN Vf=3.0 If=20mA
BLUE Vf=2.5~3.2

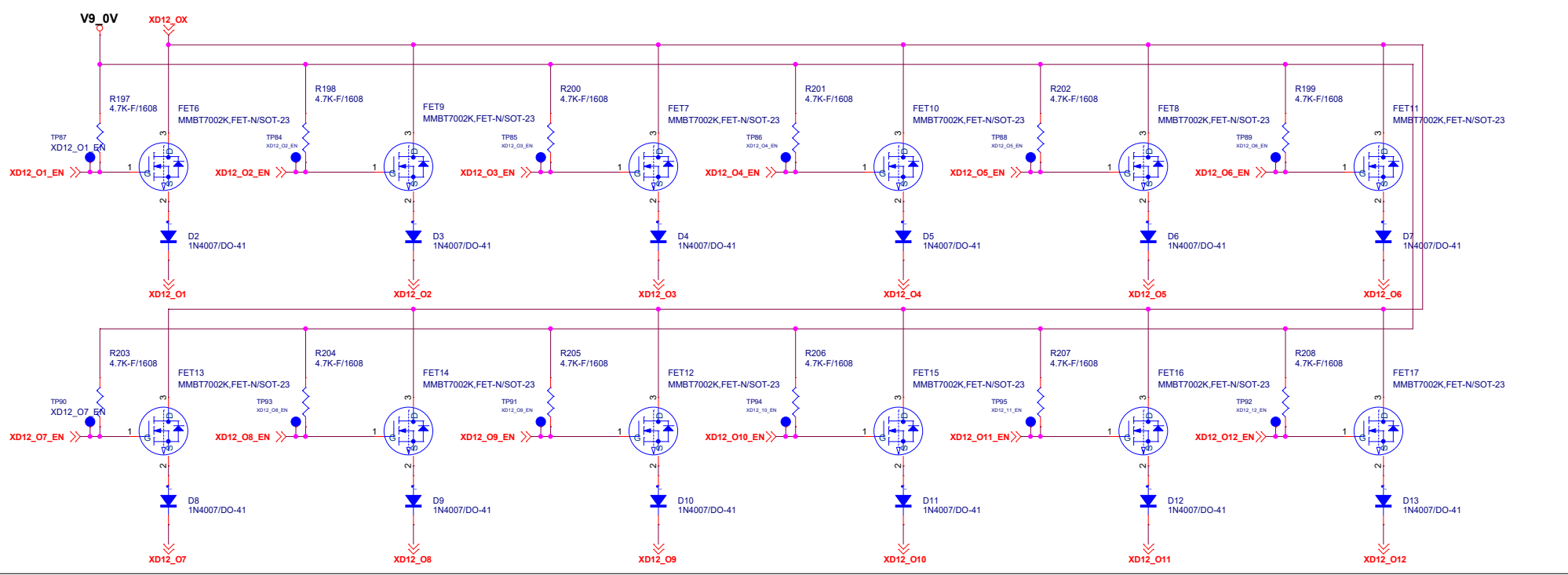


Title			
XD12R, XC24R TRIM JIG			
Size	Document Number	Rev	
Custom	XD12R TEST MODE SET	1R	
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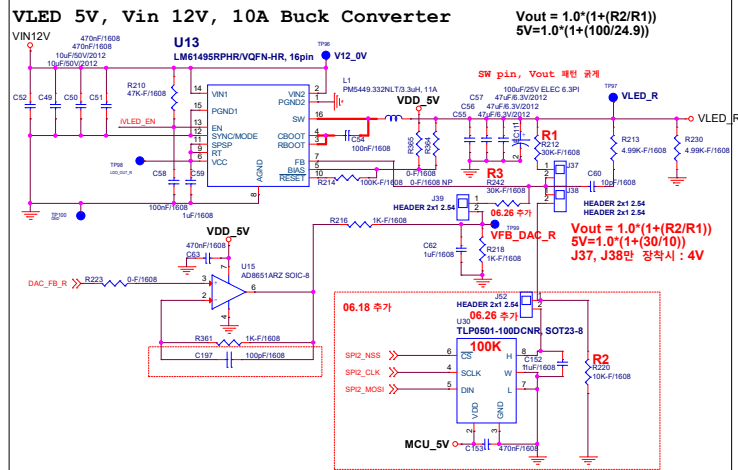
Current Sensing



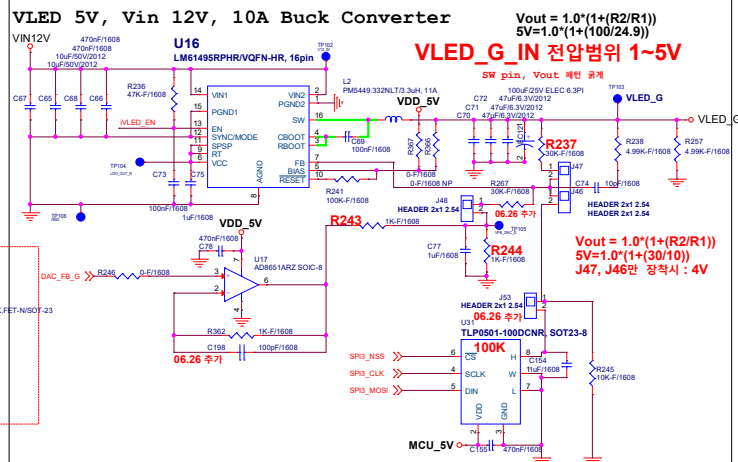
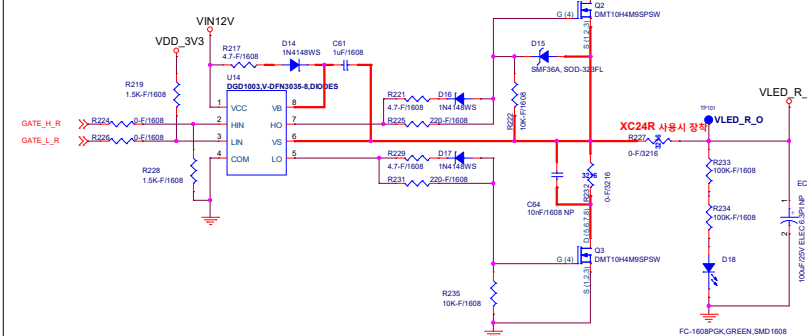
Output Select



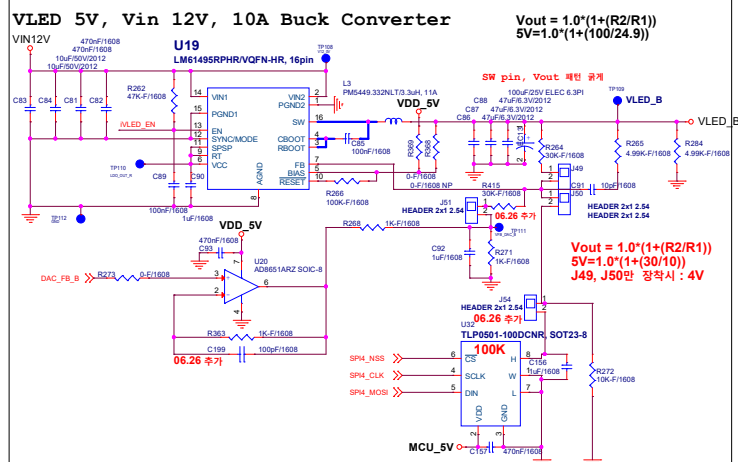
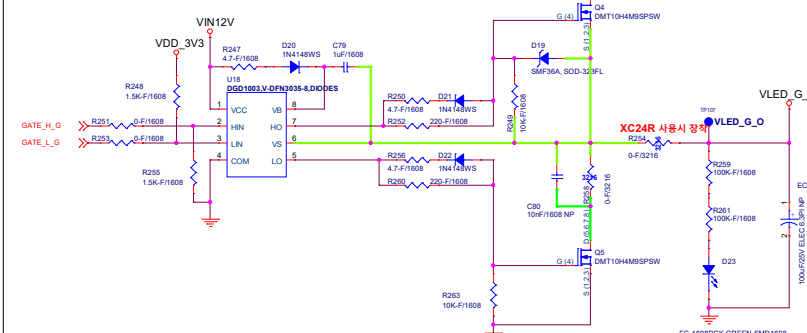
Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size	Project name XD12R, XC24R TRIM JIG	Rev	P.C.G	-
Custom	Page name CURRENT MEASURE	2.0		
	Sheet 11 of 16			
	Date: Saturday, June 27, 2026			



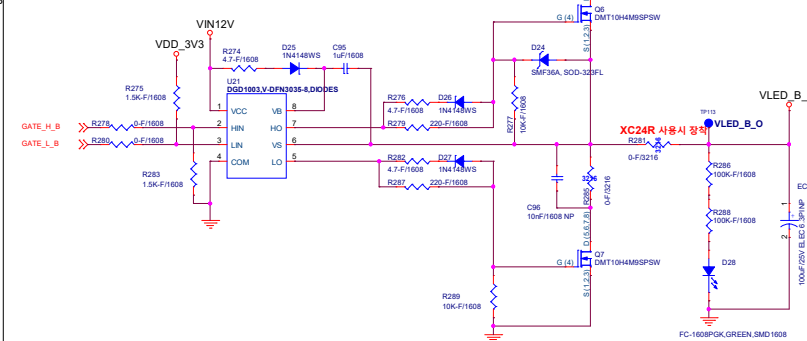
GATE DRIVER - VLED R

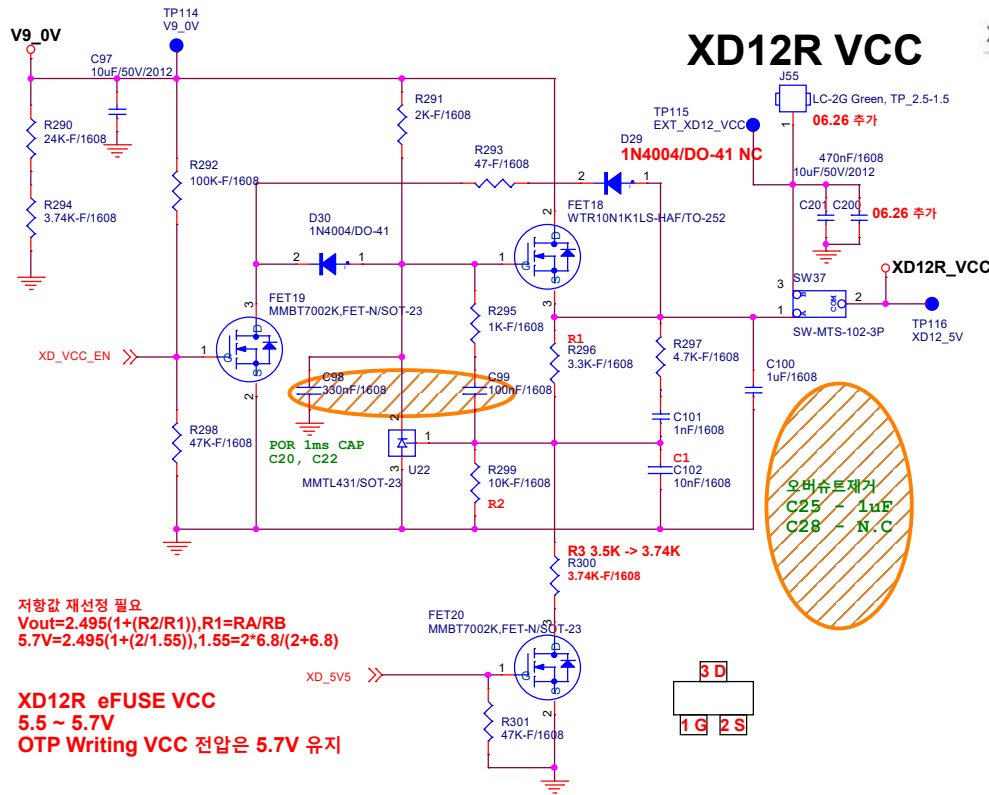


GATE DRIVER - VLED G



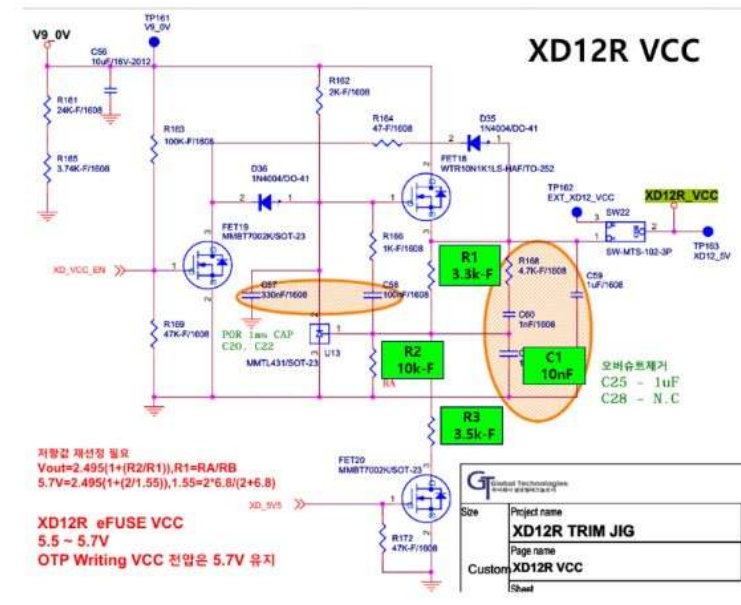
GATE DRIVER - VLED B





XD12R VCC

XD12R_VCC DC-DC Vout = 3.3V , 5.7V



XD12R VCC

Normal 조건 XD12R VCC
 $V_{out} = V_{ref} \times (1 + R1/R2)$
 $= 2.495 \times (1 + 3.3k/10k)$
 $= 3.3V$

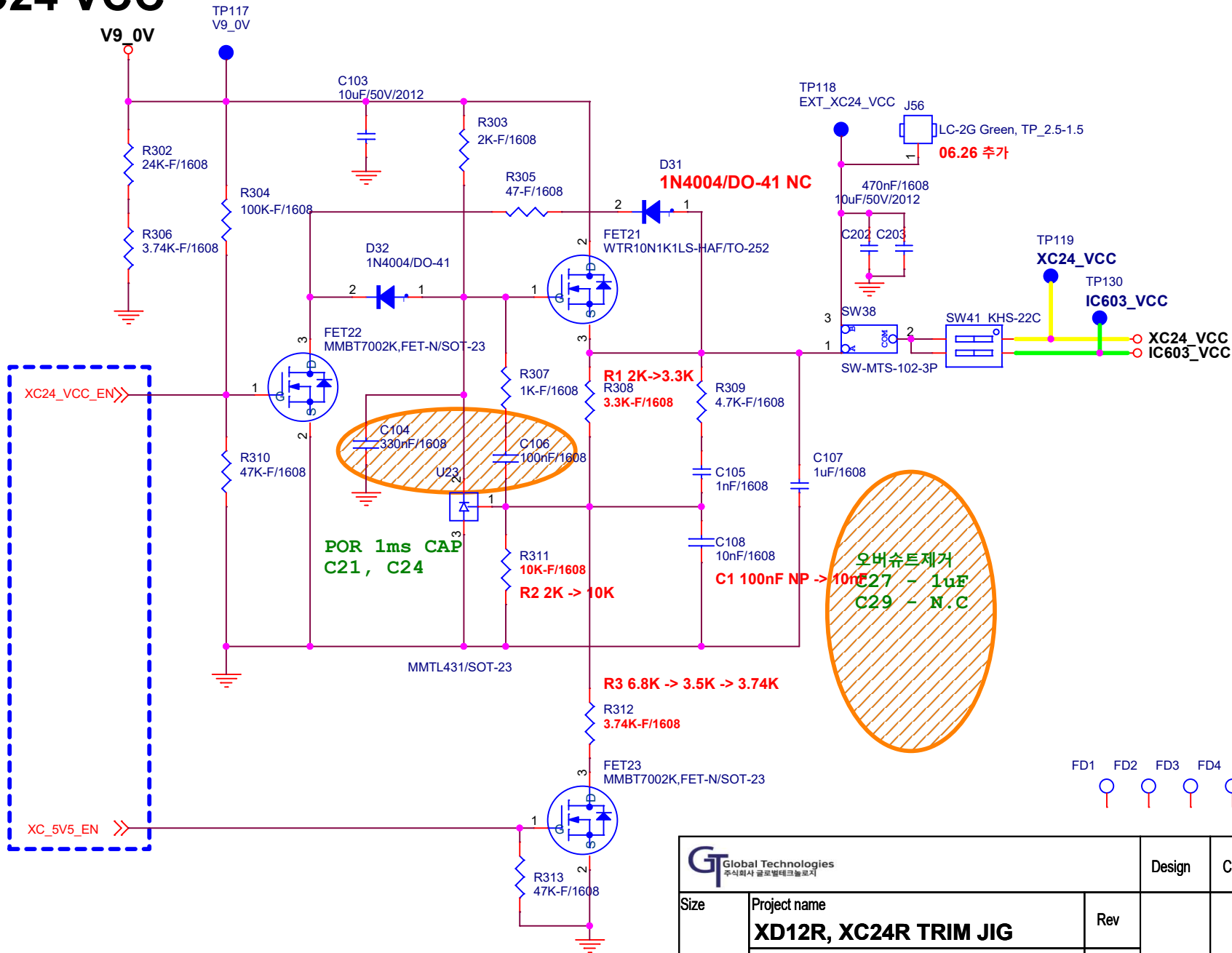
OTP Writing 조건 XD12R VCC
 $V_{out} = V_{ref} \times (1 + R1/R2)$
 $= 2.495 \times (1 + 3.3k/R2_{new})$
 $= 2.495 \times (1 + 3.3k/2.57k)$
 $= 5.7V$

R2_new
 $V_{out} = 5.7V$
 $R1/R2_{new} \approx 1.285$
 $R2_{new} = 3.3k / 1.285 = 2.57k$

R2_new
 $R2_{new} = R2 \parallel R3$
 $2.57k = 10k \parallel R3$
 $= (10k \times R3) / (10k + R3)$
 $R3 \approx 3.5k$

 Global Technologies 全球資訊科技集團有限公司			Design	Check
Size	Project name XD12R, XC24R TRIM JIG	Rev	P.C.G	-
Custom	Page name XD12R VCC	1R0		
	Sheet 13 of 16			
	Date: Saturday, June 27, 2026			

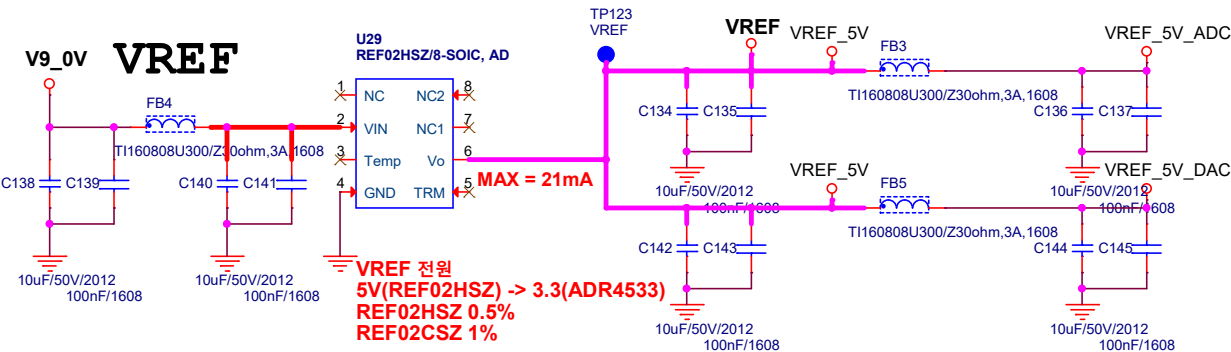
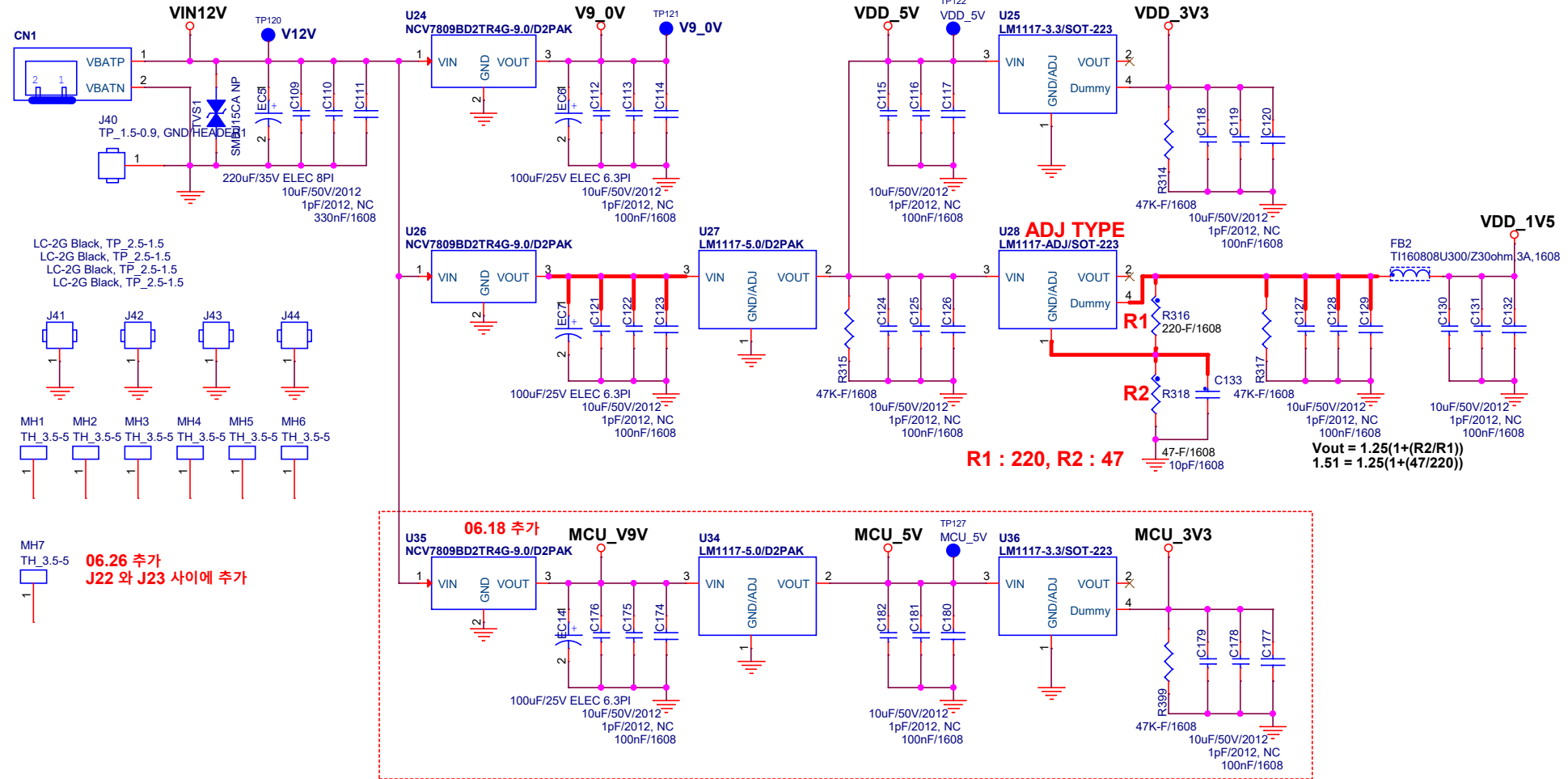
XC24 VCC



GT Global Technologies 주식회사 글로벌테크놀로지			Design	Check
Size	Project name	Rev	P.C.G	-
Custom	XD12R, XC24R TRIM JIG			
	Page name	2.0		
	XC24R POWER			
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LOGIC POWER STANDBY

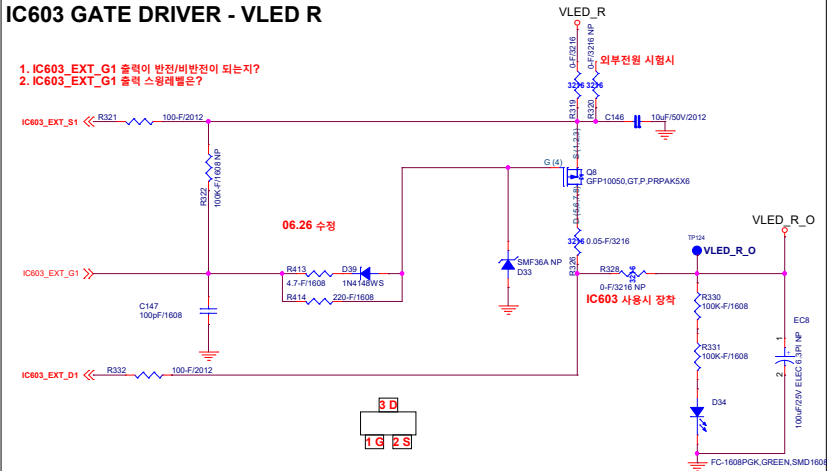
YW396-02V/1R*2-pin, 3.96mm, ST, #16~24, 7.96A, DIP, Mate YH396-02V



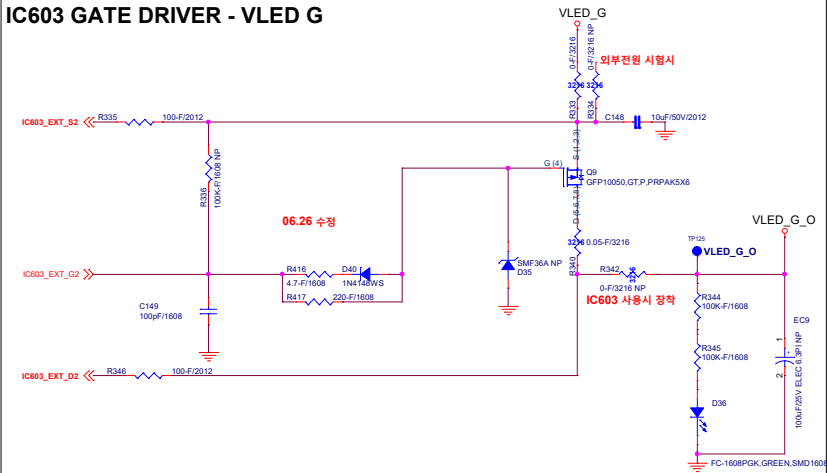
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Size	Project name XD12R, XC24R TRIM JIG	Rev	P.C.G	-
Custom	Page name LOGIC POWER	1R0		
	Sheet 15 of 16			
	Date: Saturday, June 27, 2026			

IC603 GATE DRIVER - VLED R

- 1. IC603_EXT_G1 솔라이 반전/비반전이 되는지?
- 2. IC603_EXT_G1 솔라이 스위칭레벨은?



IC603 GATE DRIVER - VLED G



IC603 GATE DRIVER - VLED B

